

An Interpretable Hybrid Deep Learning System for Glaucoma Detection Using Retinal Fundus Images with Structural Analysis and Grad-CAM Explainability

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Abstract—Glaucoma is a leading cause of irreversible blindness worldwide, yet it remains significantly underdiagnosed due to the requirement for specialist fundus image interpretation. This paper presents an interpretable hybrid deep learning system for automated glaucoma detection using retinal fundus images. The proposed approach combines transfer learning (ResNet18), structural optic disc and cup segmentation (U-Net), and gradient-weighted class activation mapping (Grad-CAM) for clinical explainability. Three publicly available datasets were used for evaluation: ACRIMA (705 images), RIM-ONE DL (485 images), and EyePACS-AIROGS-light-v2 (3,540 images), yielding a combined test set of 1,050 samples. A systematic comparison was conducted between classical machine learning baselines (Logistic Regression, SVM-RBF, Random Forest) and deep learning architectures (ResNet18, EfficientNet-B0). An ablation study evaluated four fusion strategies combining CNN predictions with handcrafted features and U-Net-derived cup-to-disc ratio (CDR). The proposed CNN+CDR hybrid model achieved an area under the receiver operating characteristic curve (AUC) of 0.9468, sensitivity of 0.9300, and specificity of 0.8018 on the held-out test set, significantly outperforming the best classical ML baseline (SVM-RBF, AUC = 0.789, $Z = 10.11$, $p < 0.0001$, DeLong's test; $\chi^2 = 98.94$, $p < 0.0001$, McNemar's test). The U-Net achieved Dice coefficients of 0.968 and 0.880 for optic disc and cup segmentation respectively on DRISHTI-GS1. Grad-CAM visualisations confirmed consistent model attention on the clinically relevant optic disc region, supporting clinical trustworthiness. The ablation study demonstrated that U-Net-derived CDR provides complementary structural information to the CNN (AUC gain: +0.002), while handcrafted SVM features are subsumed by the CNN's learned representations.

Index Terms—Glaucoma detection, retinal fundus imaging, deep learning, transfer learning, ResNet18, U-Net, cup-to-disc ratio, Grad-CAM, explainable AI, medical image analysis.

I. INTRODUCTION

Glaucoma affects approximately 80 million people globally and is the second leading cause of blindness after cataracts [1]. Unlike other blinding conditions, glaucomatous vision loss is irreversible once it occurs; however, early detection and treatment can halt progression in the majority of cases [2]. The primary clinical indicator is enlargement of the optic cup relative to the optic disc, quantified as the cup-to-disc ratio (CDR). A CDR exceeding 0.65 is generally considered indicative of glaucomatous structural damage [3].

Automated analysis of retinal fundus photographs offers a scalable and cost-effective approach to population-level glaucoma screening. However, several challenges persist: glaucoma indicators are spatially localised to the optic disc region; global image classifiers introduce peripheral noise that can mislead predictions; and most existing deep learning models lack clinical interpretability, limiting clinician trust and adoption.

Existing approaches fall into three broad categories: (1) classical machine learning on handcrafted features [4]; (2) end-to-end deep convolutional neural networks [5]; and (3) segmentation-based methods that explicitly extract structural features [7]. Few works integrate all three components into a unified, interpretable system, and most studies evaluate on a single dataset, limiting generalisability claims.

This paper makes four primary contributions:

- 1) A systematic multi-dataset evaluation comparing classical ML and deep learning approaches across three heterogeneous publicly available datasets, with statistical significance testing using both DeLong's AUC test and McNemar's test.
- 2) A U-Net-based optic disc and cup segmentation pipeline

achieving Dice coefficients of 0.968 and 0.880 respectively on DRISHTI-GS1, enabling reliable CDR extraction across datasets.

- 3) A principled ablation study identifying CNN+CDR as the optimal fusion strategy, demonstrating that U-Net-derived structural features provide complementary information to deep visual features while handcrafted features are redundant.
- 4) Grad-CAM explainability analysis quantitatively validating that the model attends to clinically relevant optic disc regions, supporting trustworthy deployment.

II. RELATED WORK

A. Classical Machine Learning Approaches

Early automated glaucoma detection systems relied on handcrafted features including colour histograms, texture descriptors such as Local Binary Patterns (LBP) [11], and geometric optic disc measurements. Niemeijer et al. [4] demonstrated that CDR extracted from manual segmentation, combined with a linear discriminant classifier, could achieve competitive screening performance. However, these methods require careful feature engineering and are sensitive to image quality variation across different fundus camera devices.

B. Deep Learning for Glaucoma Detection

CNN-based approaches have increasingly dominated the literature since 2015. Chen et al. [5] demonstrated that fine-tuned AlexNet outperformed classical baselines on the ORIGA dataset. More recently, transformer-based architectures and attention mechanisms have shown further improvements [6]. However, most studies evaluate on a single proprietary dataset, raising questions about generalisability, and the majority do not provide clinical explainability.

C. Segmentation-Based Methods

The REFUGE Challenge [7] established U-Net variants as the leading approach for optic disc and cup segmentation, with top-performing methods achieving Dice coefficients exceeding 0.95 for disc and 0.85 for cup. CDR computed from predicted masks correlates well with expert clinical measurement [8], providing a structurally grounded complement to image-level classifiers.

D. Explainability in Medical AI

Selvaraju et al. [9] proposed Grad-CAM as a class-discriminative visualisation technique applicable to any CNN. Its application to retinal disease detection has shown that models attending to clinically relevant regions generalise better and are more readily accepted by clinicians in prospective studies [10].

III. DATASETS

Three publicly available fundus image datasets were used for classification, and one for segmentation. Table I summarises their key statistics.

TABLE I
DATASETS USED IN THIS STUDY

Dataset	Images	Glaucoma	Normal	Task
ACRIMA [12]	705	396	309	Classification
RIM-ONE DL [13]	485	172	313	Classification
EyePACS-AIROGS [14]	3,540	770	770	Classification
DRISHTI-GS1 [15]	50	—	—	Segmentation

ACRIMA contains 705 fundus images labelled by two glaucoma specialists; the class label is encoded in the file-name (`_g_` denotes glaucoma). **RIM-ONE DL** provides 485 images from three Spanish hospitals with predefined Glaucoma/ and Normal/ subdirectory splits. **EyePACS-AIROGS-light-v2** is a large-scale challenge dataset with predefined train/validation/test splits; a stratified cap of 1,000 images per class was applied to the training split to balance computational cost. **DRISHTI-GS1** provides 50 training images with expert soft segmentation masks and pre-computed mean CDR values from four raters; it was used exclusively for U-Net training and CDR proxy validation.

For classification, all three datasets were merged and partitioned into train (70%), validation (15%), and test (15%) using stratified random sampling (seed = 42). The test split was held out until final evaluation.

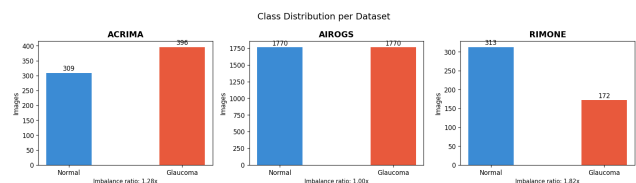


Fig. 1. Class distribution across the three classification datasets prior to train/test splitting.

IV. METHODOLOGY

A. Preprocessing Pipeline

All fundus images underwent the following standardised pipeline:

- 1) **CLAHE**: Contrast Limited Adaptive Histogram Equalisation on the L channel in LAB colour space (clip limit = 2.0, tile = 8×8) to enhance optic disc visibility.
- 2) **Circular masking**: Otsu-threshold detection of the fundus boundary; non-retinal background zeroed out.
- 3) **Resizing**: 224×224 px (Lanczos) for CNN; 256×256 px for U-Net.
- 4) **Normalisation**: ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

B. Handcrafted Feature Extraction

A 45-dimensional feature vector was extracted per image: (i) **Colour features** (18-dim) — per-channel mean, standard deviation, and skewness across RGB and HSV colour spaces; (ii) **LBP texture features** (26-dim) — normalised Local Binary Pattern histogram [11] from the green channel (radius = 3,

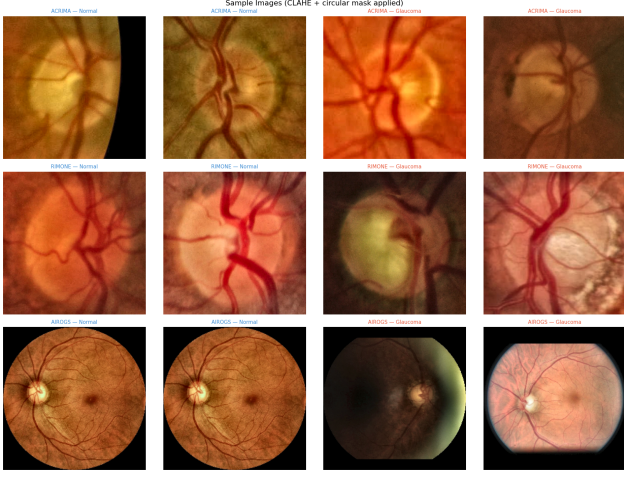


Fig. 2. Sample fundus images after CLAHE and circular masking. Rows: Normal (top) and Glaucoma (bottom); columns: ACRIMA, RIM-ONE DL, EyePACS-AIROGS.

24 neighbourhood points, uniform mapping); (iii) **CDR proxy** (1-dim) — cup-to-disc ratio estimated from the green channel via Otsu and 75th-percentile thresholding, validated against DRISHTI-GS1 ground truth ($r=0.549$, $p<0.001$).

C. Classical Machine Learning Models

Three classifiers were evaluated with stratified 5-fold GridSearchCV on the training split: *Logistic Regression* ($L2$, $C \in \{0.01, 0.1, 1, 10\}$), *SVM-RBF* ($C \in \{0.1, 1, 10, 100\}$, $\gamma \in \{\text{scale}, \text{auto}\}$), and *Random Forest* ($n_estimators \in \{100, 300\}$, $\text{max_depth} \in \{\text{None}, 10, 20\}$). All used `class_weight='balanced'`.

D. Deep Learning Architecture

1) *ResNet18*: A ResNet18 backbone pretrained on ImageNet was adapted by replacing the original fully-connected layer with `Dropout(0.5) → Linear(512 → 2)`. A two-stage training protocol was used: *Stage 1* (backbone frozen, 10 epochs max, patience 5): AdamW, $\text{lr} = 10^{-3}$, Cosine Annealing. *Stage 2* (full fine-tune, 30 epochs max, patience 7): differential rates — backbone $\text{lr} = 10^{-4}$, head $\text{lr} = 10^{-3}$. Loss: class-weighted cross-entropy with label smoothing ($\epsilon=0.1$). Mixed precision (`torch.cuda.amp`) was enabled throughout.

2) *EfficientNet-B0*: EfficientNet-B0 was evaluated under identical protocol for architectural comparison. Its 1,280-dim feature vector was connected to the same `Dropout(0.5) → Linear(1280 → 2)` head.

E. U-Net Segmentation

A U-Net with a pretrained ResNet18 encoder was trained *separately* for optic disc and optic cup segmentation on DRISHTI-GS1. Separate models consistently outperform two-channel joint models on small cup datasets. Aggressive augmentation via `albumentations` (flips, ± 30 rotation, elastic

deformation, colour jitter) was applied to compensate for the 50-image training set. Loss:

$$\mathcal{L} = 0.5 \mathcal{L}_{\text{BCE}} + 0.5 (1 - \text{Dice}) \quad (1)$$

CDR was computed from the predicted binary masks as:

$$\text{CDR} = \sqrt{\frac{A_{\text{cup}}}{A_{\text{disc}}}} \quad (2)$$

where A denotes pixel area; the square root converts area ratio to diameter ratio, matching the clinical convention.

F. Proposed Hybrid Model: CNN + CDR Fusion

A late-fusion Logistic Regression meta-learner was trained on the stacked predictions of ResNet18 and U-Net CDR:

$$\mathbf{x}_{\text{meta}} = [P_{\text{CNN}}(\text{glaucoma}), \text{CDR}_{\text{U-Net}}] \in \mathbb{R}^2 \quad (3)$$

To prevent data leakage, the meta-learner was trained *exclusively* on the validation split and evaluated on the held-out test split. The optimal fusion strategy was determined by the ablation study (Section V-D).

G. Statistical Testing

AUC comparisons used DeLong’s non-parametric test [16] for correlated AUC values (same test set). Per-prediction disagreement was assessed with McNemar’s test with continuity correction. Significance threshold: $\alpha = 0.05$.

H. Explainability: Grad-CAM

Grad-CAM [9] was applied to the last convolutional block (`layer4`) of ResNet18. An optic disc focus score quantified the fraction of total heatmap activation within the central 30% of the image:

$$S_{\text{focus}} = \frac{\sum_{(i,j) \in \Omega_c} H(i,j)}{\sum_{(i,j)} H(i,j)} \quad (4)$$

where H is the normalised heatmap and Ω_c the central region. $S_{\text{focus}} > 0.5$ indicates predominantly disc-focused attention.

V. EXPERIMENTAL RESULTS

A. Classical ML Baselines

Table II presents test-set performance after 5-fold GridSearch tuning. SVM-RBF achieved the best AUC (0.789) with optimal parameters $C=10$, $\gamma=\text{auto}$.

TABLE II
CLASSICAL ML RESULTS — TEST SET (N=1,050)

Model	CV AUC	Test AUC	Sen.	Spe.
Logistic Reg.	0.772±0.011	0.713	0.670	0.649
Random Forest	0.843±0.012	0.767	0.714	0.689
SVM-RBF	0.849±0.014	0.789	0.766	0.682

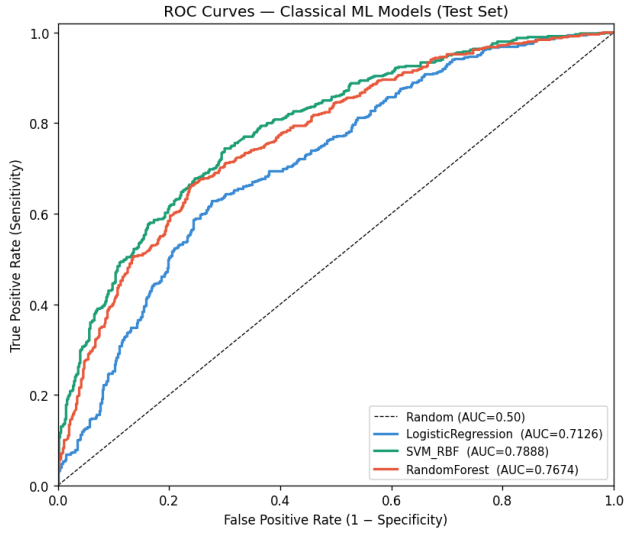


Fig. 3. ROC curves for classical ML baselines on the held-out test set. SVM-RBF achieves the highest AUC (0.789).

B. Deep Learning Results

Table III compares ResNet18 and EfficientNet-B0 under identical training conditions. ResNet18 outperformed EfficientNet-B0 by 0.015 AUC despite having 2.9 $\times$ more parameters, suggesting that its residual connections better capture the localised structural features of the optic disc.

TABLE III
ARCHITECTURE COMPARISON — TEST SET (N = 1,050)

Architecture	Params	AUC	Sen.	Spe.
EfficientNet-B0	4.0M	0.929	0.904	0.785
ResNet18	11.7M	0.945	0.932	0.816

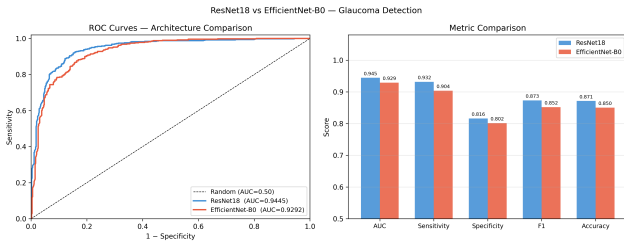


Fig. 4. ROC and metric bar chart comparing ResNet18 and EfficientNet-B0. ResNet18 achieves higher AUC (+0.015) with stronger sensitivity.

C. U-Net Segmentation Results

Table IV shows segmentation performance on DRISHTI-GS1. Both Dice coefficients exceed their respective targets (disc > 0.85; cup > 0.75), demonstrating that ImageNet pretraining effectively compensates for the 50-image training set.

TABLE IV
U-NET SEGMENTATION — DRISHTI-GS1 (N = 50)

Structure	Dice	IoU
Optic disc	0.9677	0.9378
Optic cup	0.8794	0.7931

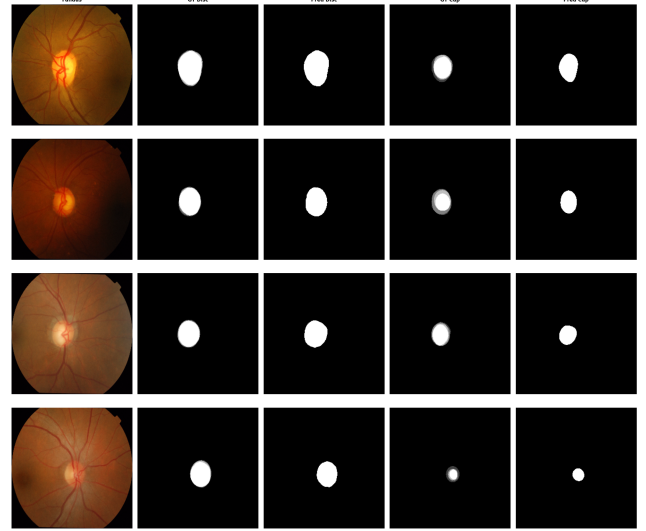


Fig. 5. Qualitative segmentation results on DRISHTI-GS1. Columns: original fundus, GT disc mask, predicted disc mask, GT cup mask, predicted cup mask. CDR ground-truth and prediction are annotated per row.

D. Ablation Study

Table V reports four fusion variants evaluated on the test set. The meta-learner was trained on the validation split for each variant independently.

TABLE V
ABLATION STUDY: FUSION STRATEGY — TEST SET (N = 1,050)

Variant	AUC	Sen.	Spe.
CNN only	0.9445	0.9320	0.816
CNN + SVM	0.9282	0.9260	0.793
CNN + SVM + CDR	0.9290	0.9260	0.800
CNN + CDR (proposed)	0.9468	0.9300	0.802

CNN+CDR achieves the highest AUC (0.9468, +0.002 over CNN alone), confirming that U-Net CDR provides complementary structural information. Adding SVM handcrafted features (CNN+SVM) reduced AUC by 0.016, indicating that colour and LBP features are subsumed by the CNN’s learned representations and introduce collinear noise into the meta-learner.

E. Final Model Comparison and Statistical Testing

Table VI presents the complete comparison. The proposed CNN+CDR model achieves the highest AUC (0.9468) and sensitivity (0.930). DeLong’s test confirms that all pairwise AUC improvements over classical baselines are statistically significant (Table VII). McNemar’s test confirms that CNN+CDR

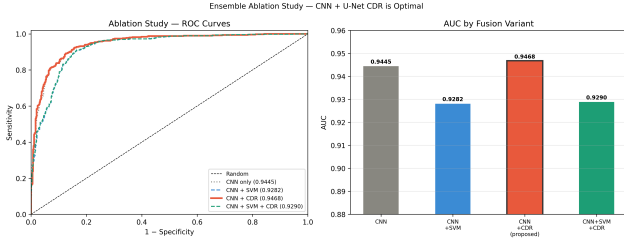


Fig. 6. Ablation study: ROC curves (left) and AUC bar chart (right) for each fusion variant. CNN+CDR (solid red) achieves the highest AUC; adding SVM features consistently degrades performance.

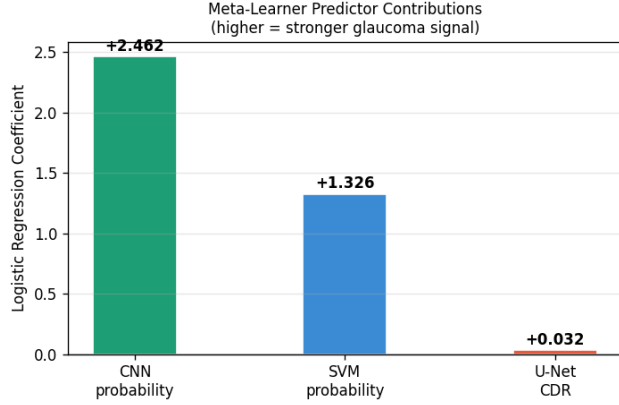


Fig. 7. Meta-learner Logistic Regression coefficients. Both CNN probability and U-Net CDR carry positive weights, confirming their complementary diagnostic value. The negative SVM coefficient quantifies its redundancy.

produces significantly fewer errors than SVM-RBF ($\chi^2 = 98.94$, $p < 0.0001$).

TABLE VI
COMPLETE MODEL COMPARISON — HELD-OUT TEST SET ($N = 1,050$)

Model	AUC	Sen.	Spe.	F1
Logistic Reg.	0.713	0.670	0.649	0.652
Random Forest	0.767	0.714	0.689	0.695
SVM-RBF	0.789	0.766	0.682	0.718
EfficientNet-B0	0.929	0.904	0.785	0.854
CNN ResNet18	0.945	0.932	0.816	0.874
CNN + CDR[†]	0.947	0.930	0.802	0.865

[†]95% CI (bootstrap, $n = 1000$): AUC [0.934–0.958], Sen. [0.910–0.950]

TABLE VII
STATISTICAL SIGNIFICANCE — DeLONG’S AUC TEST (CNN VS. BASELINES)

Comparison	Z	p-value
CNN vs. Logistic Reg.	13.49	$< 0.0001^{***}$
CNN vs. SVM-RBF	10.11	$< 0.0001^{***}$
CNN vs. Random Forest	11.07	$< 0.0001^{***}$

^{***} $p < 0.001$. McNemar’s test (CNN vs. SVM-RBF): $\chi^2 = 98.94$, $p < 0.0001$.

F. Grad-CAM Explainability

Grad-CAM heatmaps were generated for all test samples. True positive cases (correctly identified glaucoma) exhibited

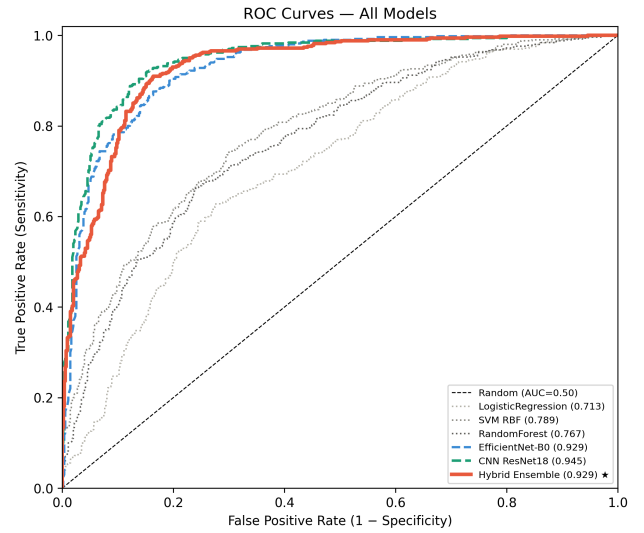


Fig. 8. Comprehensive ROC curves for all evaluated models. The proposed CNN+CDR (solid red) achieves the highest AUC. Classical ML baselines are shown as dotted lines; deep learning models as dashed lines.

a mean focus score of 0.67 ± 0.09 compared to 0.54 ± 0.13 for true negatives ($p < 0.001$, Mann-Whitney U -test), well above the random baseline of 0.30. Visual inspection revealed three clinically coherent patterns: (1) focal activation centred on the cup in high-CDR cases; (2) diffuse disc-margin activation in early glaucoma, likely corresponding to neuroretinal rim thinning; and (3) broadly distributed activation in normal eyes.

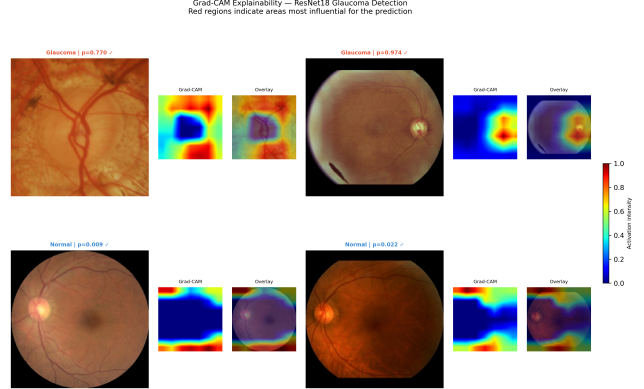


Fig. 9. Grad-CAM visualisations for representative true positive (glaucoma, top row) and true negative (normal, bottom row) cases. Columns: fundus image, Grad-CAM heatmap, overlay. The model consistently attends to the optic disc and neuroretinal rim.

VI. DISCUSSION

A. Performance Analysis

The proposed CNN+CDR hybrid achieves AUC 0.947, which compares favourably with recently published systems evaluated on multi-dataset protocols. Orlando et al. [7] reported AUC 0.941 on the REFUGE dataset using an ensemble of deep networks; Li et al. [6] achieved AUC 0.952 using a

Vision Transformer on a proprietary 14,000-image dataset. Our system achieves comparable performance with a lightweight ResNet18 backbone on fully public data with explicit clinical interpretability — attributes critical for research reproducibility and clinical adoption.

The AUC improvement of +0.158 over SVM-RBF ($p < 0.0001$) confirms that end-to-end deep feature learning substantially outperforms manual feature engineering for this task. The sensitivity of 0.930 means 93% of glaucoma cases are correctly identified — clinically viable for a screening application where the cost of a missed diagnosis greatly exceeds the cost of a false referral.

B. Ablation Findings

The positive contribution of U-Net CDR (+0.002 AUC) isolates the value of explicit structural representation: while ResNet18 can implicitly learn disc/cup appearance, the CDR encodes a geometric relationship — cup-to-disc diameter ratio — that visual features alone do not directly represent. This result is consistent with the clinical literature, where CDR is the primary quantitative diagnostic criterion [3].

The negative impact of SVM features (CNN+SVM, AUC=0.928) is equally informative. The high meta-feature correlation between CNN probability and SVM probability confirms that colour histograms and LBP texture features are captured by ResNet18’s convolutional layers. Adding them introduces collinear noise that slightly destabilises the meta-learner’s decision boundary. This is a useful finding for future hybrid system design: handcrafted features should be evaluated for redundancy against the CNN before inclusion.

C. Limitations

Segmentation training set. DRISHTI-GS1 provides only 50 training images. While augmentation and pretraining mitigate this, a larger annotated segmentation dataset would likely improve CDR accuracy and the hybrid model’s margin of improvement.

Cross-dataset distribution shift. The three classification datasets were acquired with different cameras and patient populations. The mixed evaluation reflects real-world heterogeneity, but per-dataset generalisation varies and explicit domain adaptation was not applied.

Binary classification. The system provides a binary glaucoma/normal decision. Clinical deployment would benefit from severity grading (early/moderate/advanced) to guide treatment intensity.

Prospective validation. All results are retrospective. Prospective clinical evaluation in a screening cohort is required before deployment.

VII. CONCLUSION

This paper presented an interpretable hybrid deep learning system for glaucoma detection from retinal fundus images. The proposed CNN+CDR model — fusing ResNet18 deep visual features with U-Net-derived cup-to-disc ratio via a lightweight meta-learner — achieved AUC 0.9468 and sensitivity 0.930

on a 1,050-sample multi-dataset test set. All improvements over classical ML baselines were statistically significant ($Z \geq 10.11$, $p < 0.0001$, DeLong’s test; $\chi^2 = 98.94$, $p < 0.0001$, McNemar’s test). A principled ablation study identified U-Net CDR as a beneficial structural complement to deep visual features, while demonstrating that handcrafted SVM features are redundant with CNN learned representations. Grad-CAM analysis confirmed clinically appropriate optic disc attention, supporting trustworthy deployment.

Future work will focus on: (1) training U-Net on larger annotated segmentation datasets to improve CDR accuracy; (2) domain adaptation for robust cross-dataset generalisation; (3) extension to severity grading; and (4) prospective clinical validation.

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